Computer Graphics

Lighting & Shading

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Light Source

- Any object that is emitting radiant energy that contributes to the lighting effects for other objects in a scene
- Specified by:
  - Position
  - Color of the emitted light
  - Emission direction
  - Shape

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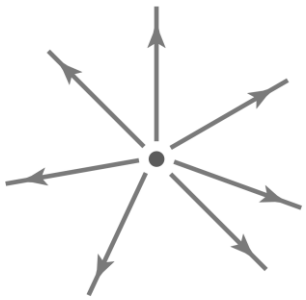
Point Light Source

Types of Light Sources



- The simplest model for an object that is emitting radiant energy
- Defined by position and colour of emitted light.
- Point light sources illuminate in all directions.
- Light rays are generated along radially diverging paths from single-colour source position.
- A reasonable approximation for sources whose dimensions are small compared to the size of objects in the scene.

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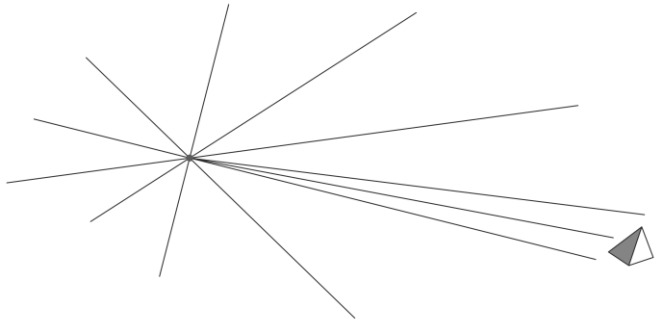
Directional Light

Types of Light Sources

- A large light source, such as the sun, that is very far from a scene can also be approximated as a point emitter, but there is little variation in its directional effects.
- Illuminates the scene from only one direction
- The light path from a distant light source to any position in the scene is nearly constant
- The vector for the emission direction and the light-source color are needed in the illumination calculations, but not the position of the source.



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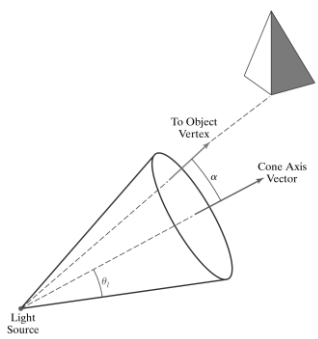
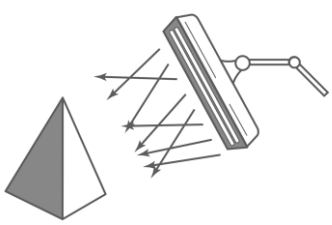
Spotlight



- A directional light source that produces a conical beam of light along the axis of cone.
- Defined by:
  - Position
  - Colour
  - Vector direction
  - Angular limit θ_l
- If an object is outside the source cone, it is not illuminated by the spot light
- Light intensity is attenuated
  - Angularly away from the cone axis vector
  - Radially out from the point-source position

Extended Light Sources



- When we want to include a large light source at a position close to the objects in a scene.
- Can be approximated as a light emitting surface with:
  - A grid of directional point emitters.
  - Set the direction for the point sources so that objects behind the light-emitting surface are not illuminated.
  - Other controls to restrict the direction of the emitted light near the edges of the source.

Shading

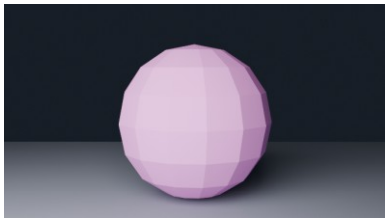
Surface Rendering

Shading for Polygons

- The illumination equation can be evaluated at every pixel to compute the colour/intensity there.
- This is expensive computationally. (Does it give the correct results?)
- Can we take advantage of the coherence or the fact that we are computing for a planar polygon?
- Can apply the illumination model to a few selected points and approximate the intensity at the other surface positions.

Constant-Intensity Surface Rendering

Types of Shading



- The simplest method for rendering a polygon surface is to assign the same color to all projected surface positions.
- Evaluate Illumination equation once per polygon
- Apply the intensity values to the whole polygon
- Each polygon gets a constant colour, giving the flat look.
- Most easy computationally.
- Which point? Centre? First vertex? Results may vary.

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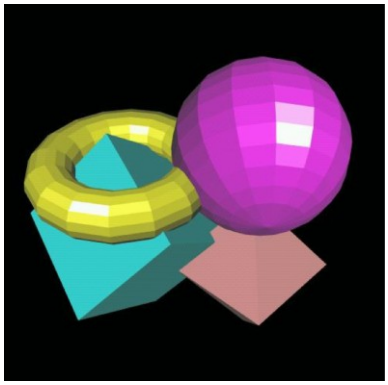
Constant-Intensity Surface Rendering

Types of Shading



Provides accurate results, if

- The polygon is one face of a polyhedron and not a section of a curved surface approximation mesh.
- All light sources illuminating the polygon are sufficiently far from the surface, such that $\mathbf{N} \cdot \mathbf{L}$ and the attenuation function are constant over the area of the polygon.
- The viewing position is sufficiently far from the polygon, such that $\mathbf{V} \cdot \mathbf{R}$ is constant over the area of the polygon.

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Gouraud Surface Rendering

Types of Shading

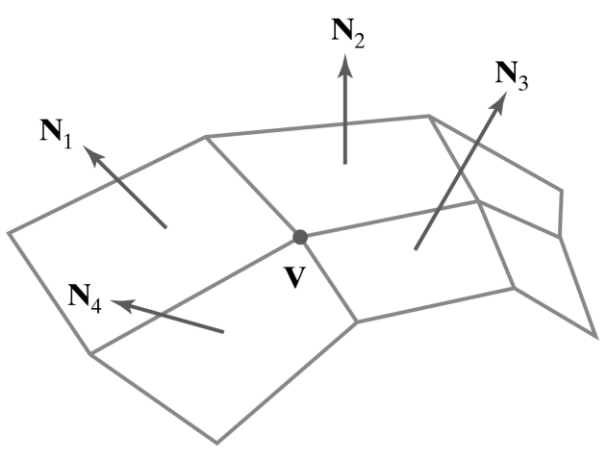
- Linearly interpolates vertex intensity values across the polygon faces of an illuminated object.
- Smoothly transitions the intensity values for each polygon facet into the values for adjacent polygons along the common edges.
- This interpolation of intensities across the polygon area eliminates the intensity discontinuities that can occur in flat surface rendering

Gouraud Surface Rendering

Types of Shading

1. Determine Vertex Normal

- At each polygon vertex, obtain a normal vector by averaging the normal vectors of all polygons in the surface mesh that share that vertex.
- $\mathbf{N}_v = \frac{\sum_{k=1}^{n} \mathbf{N}_k}{\left|\sum_{k=1}^{n} \mathbf{N}_k\right|}$
- vertex v has n adjoining faces



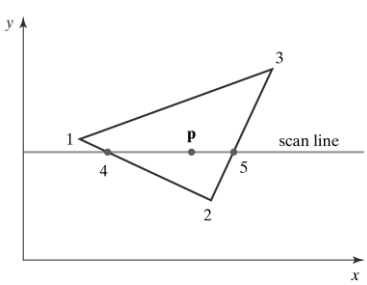
Gouraud Surface Rendering

2. Apply Illumination Model

- Apply an illumination model at each polygon vertex to obtain the light intensity at that position.

Gouraud Surface Rendering

3. Interpolate the Vertex Intensities

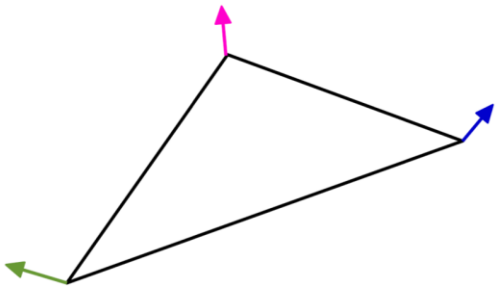


- After all vertex intensities have been computed for a polygonal facet, we can interpolate the vertex values to obtain the intensities at positions along scan lines that intersect the projected area of the polygon.
- For each scan line, the intensity at the intersection of the scan line with a polygon edge is linearly interpolated from the intensities at the endpoints of that edge.
- $I_4 = \frac{y_4 - y_2}{y_1 - y_2} I_1 + \frac{y_1 - y_4}{y_1 - y_2} I_2$

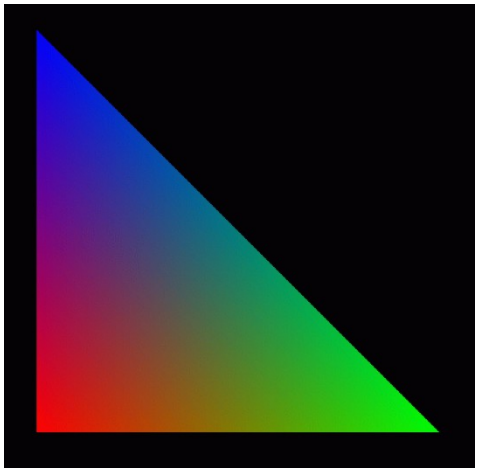
Gouraud Surface Rendering

Types of Shading

Vertex Normals



Shading



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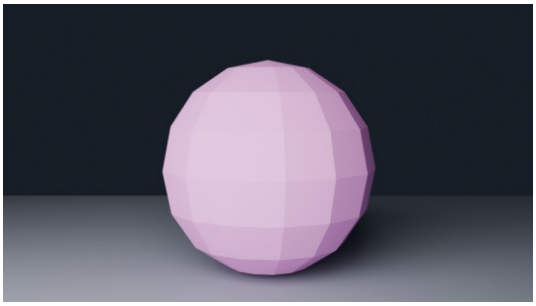
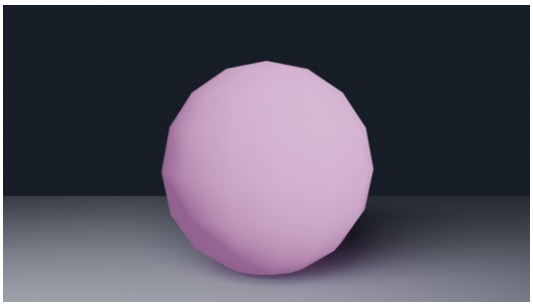
Shading Comparison

Types of Shading

Flat



Gourand



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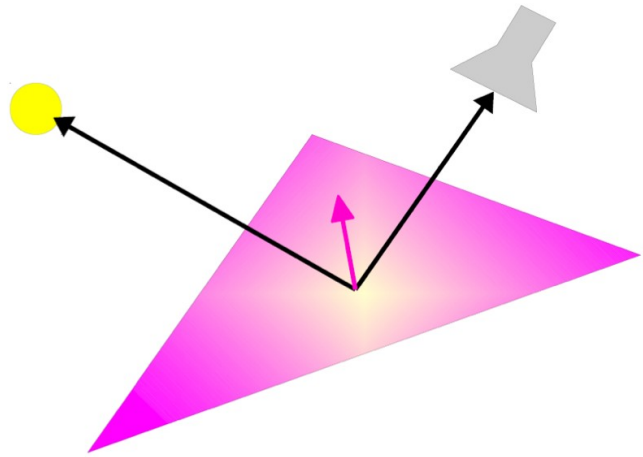
Highlights under Interpolated Shading

- Specular highlights are localized bright areas.
- If the highlight falls in the middle of a polygon, it will be completely missed as illumination equation is computed only for the vertices.
- If highlight falls on a vertex, it will be interpolated across, making it less local.
- These cannot be handled by interpolation of intensities.

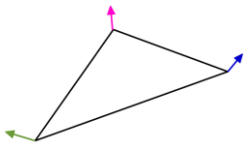


Phong Surface Rendering

Types of Shading

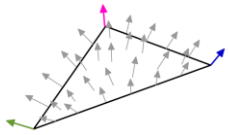
1. Determine Vertex Normal

Determine the average unit normal vector at each vertex of the polygon



2. Interpolate the vertex Normal

Interpolate the vertex normals linearly over the projected area of the polygon



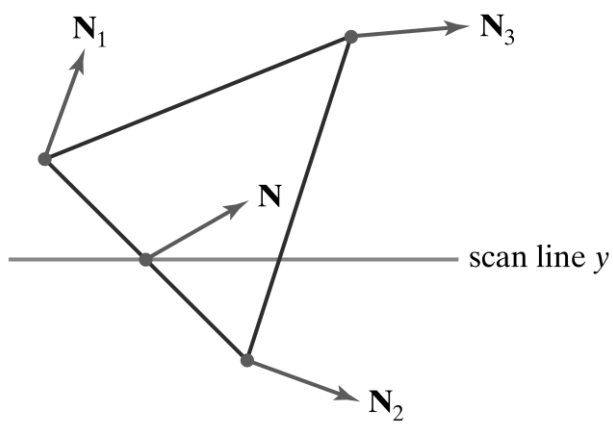
3. Apply Illumination Model

Apply an illumination model at positions along scan lines to calculate pixel intensities using the interpolated normal vectors

Phong Surface Rendering

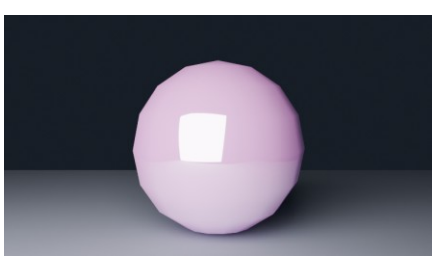
Types of Shading

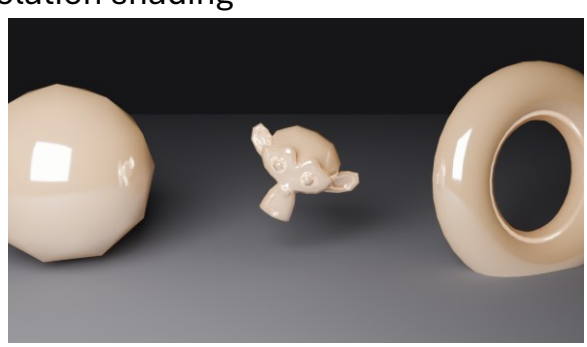
- Interpolates normal vectors instead of intensity values.
- The result is a more accurate calculation of intensity values, a more realistic display of surface highlights.
- Phong method requires more computation than the Gouraud method.



Phong Surface Rendering

- Computes highlights very well.
- Computation time is more as the equation is evaluated at every pixel.
- Also called normal vector interpolation shading

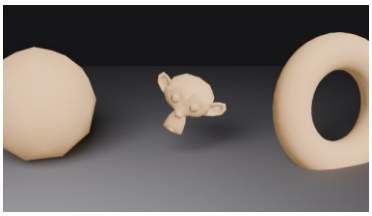




Shading Comparison

Flat

Gourard



Phong

Thank You